

Feature Engineering

Feature engineering is the process of using your domain knowledge or intuition about a problem to create new features from the original ones. This is often done by transforming or combining existing features to help the learning algorithm make more accurate predictions. The choice of features can have a huge impact on your model's performance.

An Example: Housing Prices

Let's say you're predicting house prices and have two features:

- x_1 : The width (or frontage) of the plot of land.
- x_2 : The depth of the plot of land.

Initial Model

You could start with a standard linear model using these two features:

$$f(x) = w_1x_1 + w_2x_2 + b$$

This model might work reasonably well.

Engineering a New Feature

However, you might have an intuition that the total **area** of the land is a more powerful predictor of price than its width and depth treated separately.

1. **Create a new feature:** You can define a new feature, x_3 , that represents the area.

$$x_3 = x_1 \times x_2$$

2. **Build a new model:** You can now create a more sophisticated model that includes the original features and your new, engineered feature.

$$f(x) = w_1x_1 + w_2x_2 + w_3x_3 + b$$

This new model is more flexible. It allows the learning algorithm to determine the relative importance of width, depth, and the total area, potentially leading to a much better fit for the data. By defining new features, you can often create a much more effective model.